

[illegible]

Lecture due Feb 14, 2024 20:30 IST

Definition 3.1 The **column space** of a matrix **A** is the span of its columns. The notation for it is **CS (A)**. (It is also called the **image** of **A**, and written **Im (A)**.)

Since **CS (A)** is a span, it is a vector space.

Example 3.2 Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$. Its column space is given by

$$\text{CS (A)} = \text{the span of } \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$

Using that the column vectors are all constant multiples of the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, we find a basis for the column space consisting of the single vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and find that **CS (A)** is the line $y = 2x$ in $\mathbb{R}^2$.

Example 3.3 Find a basis for the column space of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 1 & 2 & 3 \\ 0 & 0 & 9 \end{pmatrix}.$$

The column space is defined as **Span** $\left(\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -3 \\ 3 \\ 9 \end{pmatrix} \right)$. But the second vector is

a multiple of the first vector, so it is redundant. Therefore, the column space can be described more simply as the span of the first and third columns:

$$\text{CS (A)} = \text{Span} \left(\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -3 \\ 3 \\ 9 \end{pmatrix} \right).$$

These two vectors are linearly independent, so we do need both vectors in the basis for this column space.

Column space concept check I

1/1 point (graded)

What is the column space of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$? Check all that apply.

☐ **CS (A) = 0.**

☒ **CS (A) = Span $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.**

☐ **CS (A) = Span $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.**

☐ $\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$

☐ $\text{CS}(\mathbf{A}) = \mathbb{R}^2.$

☐ $\text{CS}(\mathbf{A}) = \mathbb{R}^3.$

☐ $\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$

☒ $\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} \sqrt{2}/3 \\ \sqrt{2}/3 \end{pmatrix}.$



Solution:

Because all of the columns are multiples of a single vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, we know that a basis for the column space is given by the span of this one vector:

$$\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

However, this basis is not unique. Any scalar multiple of this vector is also a basis for the column space:

$$\text{CS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \text{Span} \begin{pmatrix} \sqrt{2}/3 \\ \sqrt{2}/3 \end{pmatrix}.$$

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You have used 1 of 4 attempts

i Answers are displayed within the problem

Column space concept check II

1/1 point (graded)

If you switch two columns of a matrix $\mathbf{A}$, will the new matrix have the same column space?

☒ Yes.

☐ No.

☐ It depends on the matrix.



Solution:

Yes, the column spaces are the same. Since the column space of a matrix is the span of its columns, as long as we don't change the columns themselves, we will have a span of the same set of vectors. This span will be the same regardless of the order of the columns.

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Steps to compute a basis for **CS (A)**:

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Recall that **B** is a row echelon form of **A**. Let **C** be the *reduced* row echelon form of **A**. If

fifth column = 3 (first column) + 7 (second column)

is true for a matrix, it will remain true after any row operation.

Similarly, any linear relation between columns is preserved by row operations. So the linear relations between columns of **A** are the same as the linear relations between columns of **C**. The condition that certain numbered columns (say the first, second, and fourth) of a matrix form a basis is expressible in terms of which linear relations hold. So if certain columns form a basis for **CS (C)**, the same numbered columns will form a basis for **CS (A)**. **Warning:** these are very unlikely to be the same columns!

Recall that **B** is a row echelon form of **A**. We can obtain the reduced row echelon form **C** by performing Gauss-Jordan elimination on **B**. This process does not change the pivot locations. Thus it will be enough to show that the pivot columns of **C** form a basis of **CS (C)**. Since **C** is in reduced row echelon form, the pivot columns of **C** are the first **r** of the **m** standard basis vectors for $\mathbb{R}^m$, where **r** is the number of nonzero rows of **C**. These columns are linearly **independent**, and every other column is a linear combination of them, since the entries of **C** below the first **r** rows are all zeros. Thus the pivot columns of **C** form a basis of **CS (C)**.

(The symbol (*) after the title means that you are not required to know this proof for any exam in this class.)

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In particular,

dim CS (A) = # pivot columns of B.

Warning: Usually **CS (A) ≠ CS (B)**.

Practice with algorithm

1/1 point (graded)

Which of the following sets of vectors is a basis for the column space of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 2 \\ -1 & 1 & -2 \\ 2 & 3 & 4 \end{pmatrix}?$$

 Calculator

Hint: a row echelon form for **A** is the matrix

$$\mathbf{B} = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

☐ $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$

☐ $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$

☐ $\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$



Solution:

The first and the second columns are the pivot columns in the row echelon form of **A**. Therefore, the first and the second columns $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$ of **A** form a basis for its column space.

The algorithm gives one basis of **CS (A)**, but of course it has many other bases. Let's check that no other option gives such a basis. The first option is wrong since the vectors spanned by that basis have third coordinate **0**, so we can't get any of the columns of **A** in their span. Note here that pivot columns of **B** aren't related to **CS (A)** themselves, and only point to columns of **A** that span **CS (A)**.

All bases of **CS (A)** should have the same number of vectors. The basis we found in our case has 2 vectors, so should have any other correct answer as well. Therefore both choices three and four are not correct because they have the wrong dimension.

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3. Column space

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